

ASHUTOSH JOSHI

Senior Software Engineer | AI-Powered Platforms · LLM/RAG Systems · Distributed Systems
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SUMMARY

Software engineer with 6+ years across AI/ML systems, platform services, and distributed systems — currently building the ML and data backbone of NASA's Pandora network: 300+ spectrometers on five continents running deep-learning models over an edge-to-cloud fleet. Hands-on building production LLM-enabled applications with retrieval-augmented generation (RAG), embeddings, semantic search, and agentic workflows; designing AI evaluation, guardrails, and observability for reliability and safety; and shipping secure, auditable cloud services (Azure, AWS, GCP) in Python and C++. Owns ambiguous problem spaces end to end and drives them to production-quality outcomes.

TECHNICAL SKILLS

Languages: Python, C++17, SQL, TypeScript/JavaScript, Java

Generative AI / LLM: LLM-enabled applications, retrieval-augmented generation (RAG), embeddings & vector search (pgvector), semantic search, multi-agent / agentic workflows, tool use / function calling, AI evaluation (evals) & guardrails, LLM observability & monitoring, prompt engineering & versioning, responsible AI practices (human-in-the-loop approval, audit trails), Anthropic & OpenAI APIs, AI-assisted engineering workflows

Cloud & Platform Engineering: Azure, AWS, GCP, cloud-native architectures, distributed systems, REST API design (FastAPI, Flask, Node.js), platform services, Docker, CI/CD pipelines, telemetry & monitoring, observability, model deployment (MLOps), GPU-accelerated computing, Linux

Machine Learning & Data Science: deep learning (CNNs, encoder-decoder), computer vision, time-series anomaly detection, model training & evaluation (precision/recall/F1, ROC-AUC), PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, PySpark

Data Engineering: ETL/ELT pipelines, data modeling, data cleaning & validation, Snowflake, PostgreSQL, Redis, AWS S3, Tableau, Power BI

Engineering Practices: system design, object-oriented design, data structures & algorithms, unit & integration testing (pytest), code reviews, debugging & performance optimization in production, auditable/traceable logging, Git/GitHub

PROFESSIONAL EXPERIENCE

Software Engineer — AI/ML Systems, SciGlob Instruments & Services (NASA GSFC — Pandora Global Network) | Columbia, MD | Nov 2024 – Present

- Design, build, and operate AI-powered platform services for a globally distributed fleet of 300+ instruments: deep-learning anomaly detection flags sensor and data irregularities on live streams, eliminating hours of daily manual monitoring.
- Built an encoder-decoder CNN (computer vision) for cloud detection and integrated it into the live sun-scan workflow, gating measurements on real-time sky conditions to raise scan quality network-wide.
- Engineered edge-to-cloud data processing for fleet health — parsing raw L0 instrument streams at the edge, scoring instrument health 0–100, and shipping daily summaries to cloud dashboards (GCS, Flask REST services, Pydantic).
- Created evaluation, measurement, and monitoring approaches for production ML: a fleet-wide observability dashboard accelerating triage across five continents, plus per-scan traceable logging making every data product auditable end to end.
- Leading a ground-up C++17 rewrite of the Blick spectral processing suite (L0 → L2): 14 modular libraries, a parity test harness validating against legacy output, and a GPU-ready acceleration framework for high-throughput compute.
- Automated data acquisition, cleaning, and feature-engineering pipelines aggregating measurements from 300+ spectrometers, delivering reproducible, analysis-ready datasets to NASA/ESA researchers.

Data Analyst & Developer, 407 Associates LLC | Laurel, MD | Apr 2024 – Nov 2024

- Built scalable Python ETL, data, and ML pipelines (PySpark, scikit-learn, TensorFlow) and managed AWS S3 / Snowflake warehouse workloads with optimized SQL (window functions, indexing), cutting operating costs by 25%.
- Transformed raw multi-source data into clean, validated datasets feeding analytics and ML pipelines; delivered Tableau and Power BI dashboards and internal web tools that retired manual spreadsheet workflows.

System Engineer, Tata Consultancy Services | Bangalore, IN | Jun 2020 – Aug 2022

- Engineered large-scale ETL pipelines for Albertsons (US grocery retail): 10M+ rows across sources with 20+ automated data-quality validation checks, cutting data-quality defects by 30%.
- Drove performance analysis of cloud applications under load (+25% throughput, -30% response time, 99.5–99.9% availability) and built CI/CD deployment and monitoring automation across 3–5 environments, reducing release time by 40% and incident MTTR by 25%.

SELECTED PROJECTS — github.com/ashutoshjoshi1

- CommonGround — Enterprise RAG Platform:** Production FastAPI backend with multimodal ingestion (OCR), pgvector semantic search with citations, prompt versioning, evaluation datasets, and audit trails for compliance-ready retrieval. *FastAPI · Next.js · pgvector · Redis · Docker*
- InCortex — Self-Learning Agentic AI Framework (open source):** Persistent vector memory with forgetting curves, gated tool use with human-in-the-loop approval, strategies competing under UCB bandits, and Brier/ECE calibration audits; REST API, 450 tests, 100% coverage. *Python · Vector Memory · REST*
- Claude TopstepX — LLM Decision Engine over a Deterministic Core:** LLM-driven decisions wrapped around a fully deterministic, immutable, heavily tested core — seven-dimension context scoring, risk-gate chain, ~93% test coverage. *Python · Anthropic API · pytest*

EDUCATION

University of Maryland, Baltimore County — M.P.S., Data Science (Machine Learning concentration) · GPA 3.89/4.00 | Aug 2022 – May 2024